

CATALINA, CA, USA

WMO: 722920

Lat: 33.405N

Lon: 118.416W

Elev: 1602

StdP: 13.86

Time Zone: -8.00 (NAP)

Period: 00-19

WBAN: 23191

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	42.8	44.4	4.1	7.2	65.6	8.5	9.0	64.7	28.1	47.8	25.6	47.5	10.7	280	0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	13.4	90.5	63.0	87.7	62.4	84.5	60.9	68.3	81.9	66.9	79.9	65.4	78.1	6.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.8	94.0	72.8	62.3	89.1	70.9	61.0	85.0	69.1	33.5	81.9	32.2	79.6	31.0	77.7	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.1	18.9	16.3	DB	40.1	97.2	1.7	3.7	38.9	99.9	37.9	102.1	37.0	104.2	35.8	106.9
			WB	33.7	70.9	1.6	1.7	32.5	72.1	31.6	73.1	30.7	74.0	29.5	75.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.5	56.3	54.8	56.4	56.9	59.4	63.4	68.5	70.1	70.1	65.4	60.6	55.3	(6)
	DBStd	9.40	7.37	7.58	7.30	7.23	7.27	7.22	7.21	7.59	8.57	8.57	7.82	6.34	(7)
	HDD50	69	16	23	11	4	0	0	0	0	0	0	2	13	(8)
	HDD65	2183	286	303	290	268	215	117	41	27	38	103	183	312	(9)
	CDD50	4262	212	156	209	212	293	402	574	624	604	479	320	177	(10)
	CDD65	898	17	16	23	26	42	68	151	186	192	117	50	10	(11)
	CDH74	6799	18	31	85	162	307	580	1145	1584	1740	891	247	9	(12)
	CDH80	2087	0	0	9	36	87	206	331	492	626	257	43	0	(13)
(14) Wind	WSAvg	7.0	7.8	8.4	8.3	8.3	7.1	5.6	5.3	5.5	6.1	6.8	7.2	7.9	(14)
(15) Precipitation	PrecAvg	11.6	2.2	2.4	1.6	0.7	0.3	0.1	0.1	0.0	0.2	0.6	1.1	2.0	(15)
	PrecMax	27.4	7.9	7.5	3.7	1.8	1.1	0.3	0.6	0.1	1.0	3.5	2.0	7.0	(16)
	PrecMin	5.1	0.2	0.0	0.5	0.2	0.0	0.0	0.0	0.0	0.2	0.0	0.2	0.0	(17)
	PrecStd	6.1	2.5	2.1	1.1	0.6	0.3	0.1	0.2	0.0	0.2	1.0	0.6	2.1	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.7	77.9	81.3	84.6	88.1	92.8	92.9	92.8	95.2	91.1	84.9	75.1	(19)
		MCWB	51.5	53.1	56.3	56.3	60.4	62.0	66.9	62.8	64.5	60.6	56.5	51.0	(20)
	2%	DB	73.3	73.9	75.7	78.8	81.7	85.7	86.9	88.3	90.2	86.2	80.3	71.8	(21)
		MCWB	50.7	51.2	53.6	55.0	57.9	61.4	64.7	63.0	63.1	58.8	54.7	49.9	(22)
	5%	DB	70.6	70.9	72.3	73.6	76.6	80.7	83.3	85.2	86.8	82.3	76.6	68.4	(23)
		MCWB	49.4	50.2	52.4	52.8	56.4	60.6	62.8	62.1	62.3	57.2	53.0	48.4	(24)
	10%	DB	67.9	67.6	68.4	69.2	72.0	75.7	80.5	82.2	83.1	79.2	73.1	65.5	(25)
		MCWB	48.5	48.8	50.7	51.9	55.1	59.2	62.3	60.9	60.5	55.6	51.6	47.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.1	58.5	59.7	61.5	63.9	67.4	70.2	70.1	69.8	65.7	60.5	58.3	(27)
		MCDB	64.6	63.9	71.2	75.4	80.4	84.9	85.7	83.5	85.6	77.6	68.1	61.7	(28)
	2%	WB	55.7	56.0	57.3	58.2	61.2	64.4	68.2	68.1	67.5	63.5	58.8	56.4	(29)
		MCDB	60.9	62.6	67.6	70.4	74.8	79.5	81.4	81.2	80.7	74.7	66.7	60.4	(30)
	5%	WB	54.0	53.9	55.5	56.2	59.4	62.7	66.6	66.3	65.7	61.7	57.1	54.6	(31)
		MCDB	60.3	62.1	65.1	66.9	70.6	75.2	78.6	79.0	79.9	72.3	64.9	59.3	(32)
	10%	WB	52.3	52.3	54.0	54.5	57.5	61.3	65.0	64.7	63.9	60.0	55.7	53.1	(33)
		MCDB	59.3	60.0	62.7	63.1	66.3	72.2	76.0	76.6	76.6	69.5	64.2	57.8	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.6	9.8	11.4	12.3	12.1	13.5	14.6	14.3	13.4	11.7	10.6	9.1	(35)
		MCDBR	12.0	13.7	15.2	17.9	17.5	18.4	17.2	15.2	16.1	14.8	13.3	12.4	(36)
	5% WB	MCWBR	7.4	8.7	9.5	10.0	10.4	11.1	11.7	11.9	10.5	9.0	7.5	7.5	(37)
		MCWBR	10.4	11.9	13.9	15.7	15.8	17.1	16.5	15.2	15.0	13.8	12.2	10.0	(38)
(40) Clear-Sky Solar Irradiance	taub		0.291	0.304	0.320	0.323	0.337	0.339	0.352	0.341	0.326	0.324	0.292	0.292	(40)
		taud	2.638	2.618	2.548	2.538	2.523	2.512	2.540	2.555	2.597	2.587	2.659	2.658	(41)
	Ebn at Noon		298	304	305	306	300	298	293	297	298	292	295	292	(42)
		Edh at Noon	23	27	31	33	34	34	33	32	29	27	23	22	(43)
(44) All-Sky Solar Radiation	RadAvg		1007	1289	1696	2071	2146	2121	2196	2115	1797	1380	1098	908	(44)
	RadStd		69	110	98	115	185	161	154	103	114	126	85	71	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (50 neighbors)	+1.15	N/A	-2.92	+1.40	N/A	N/A	N/A	-212	+414	+196

Nomenclature: See separate page